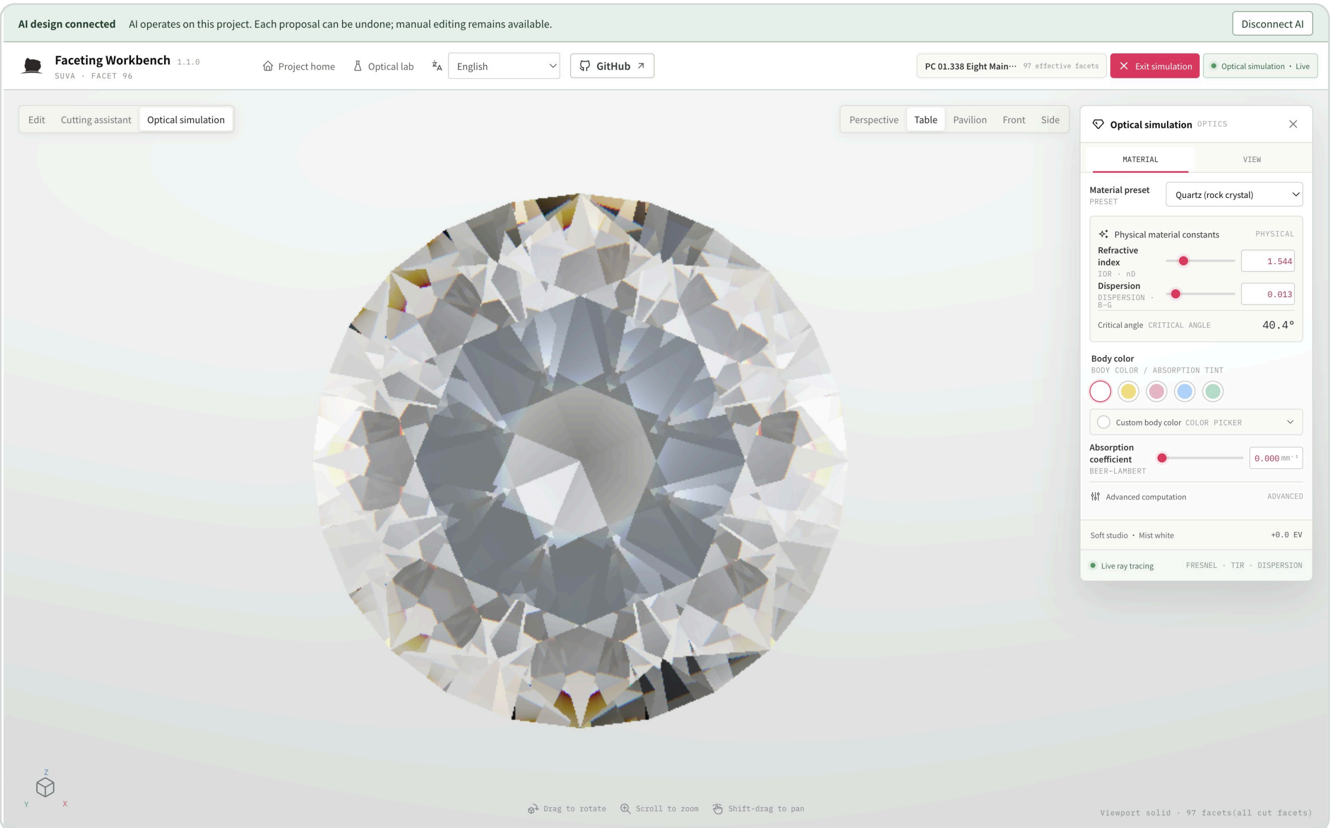
Faceting Workbench

1.1.0 · SUVA / FACET 96

Designer’s illustrated manual

From an idea to a gem of your own

Start with outline and proportion, arrange facets into a rhythm, then compare materials and lighting. Give every action a visible design purpose.



Eight Main Highlight · Actual quartz-material view of preset PC 01.338; this is not a prediction of physical fire or a machining guarantee.

Who this manual is for

For designers who adjust cuts, compare alternatives and communicate with faceters. No engineering vocabulary is required: each example starts with the shape you want to achieve.

Getting started

Prepare the exercises in docs/manual/examples and open the workbench version printed in the footer. Start with the project home on page 3, then enter the workbench using the exercise on page 4.

Reading route

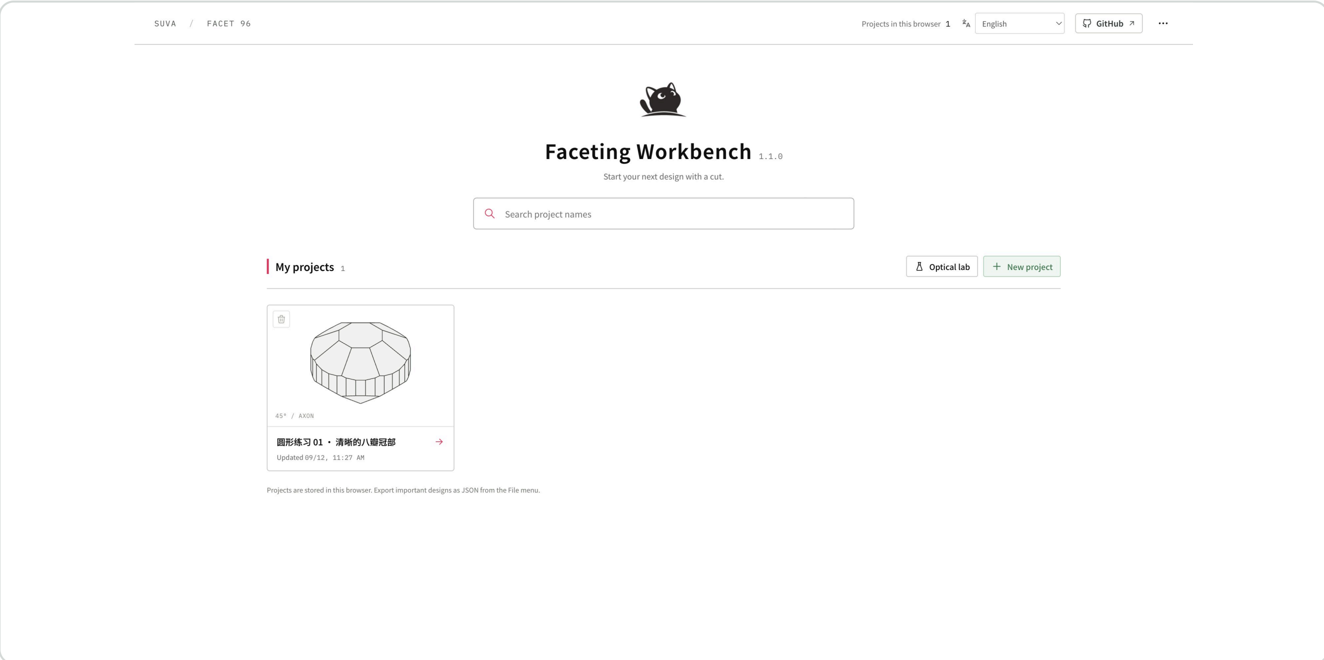
Find the design question you are working on

On your first visit, follow Start → Compare → Position → Deliver. Later, jump directly to the page that matches your design goal.

START · PROJECT HOME

Find the project you want to continue

The workbench opens at project home. Treat each project as an evolving design: name its purpose and use the actual outline on its card to recognize it.



Projects appear in order of recent updates. Cards show committed shapes; search matches project names only.

1. **Create a starting point**
   Click "New project" and choose the default start, a stock preset or an OBJ upload. The default includes a table and 32 girdle facets; shaped blanks start with no cuts.

2. **Continue a design**
   Search by name and open its card. Reopening a project saves to the same card, without making a copy. Rename the design at the top of the editor.

3. **Keep your current experiment**
   Returning home leaves the current CUT intact. "Resume design" restores that workspace; switching or creating a project asks about any unsaved preview.

Pause and make a design judgment

Can you recognize the design by its name and outline? If not, give it a short name that expresses the design goal.

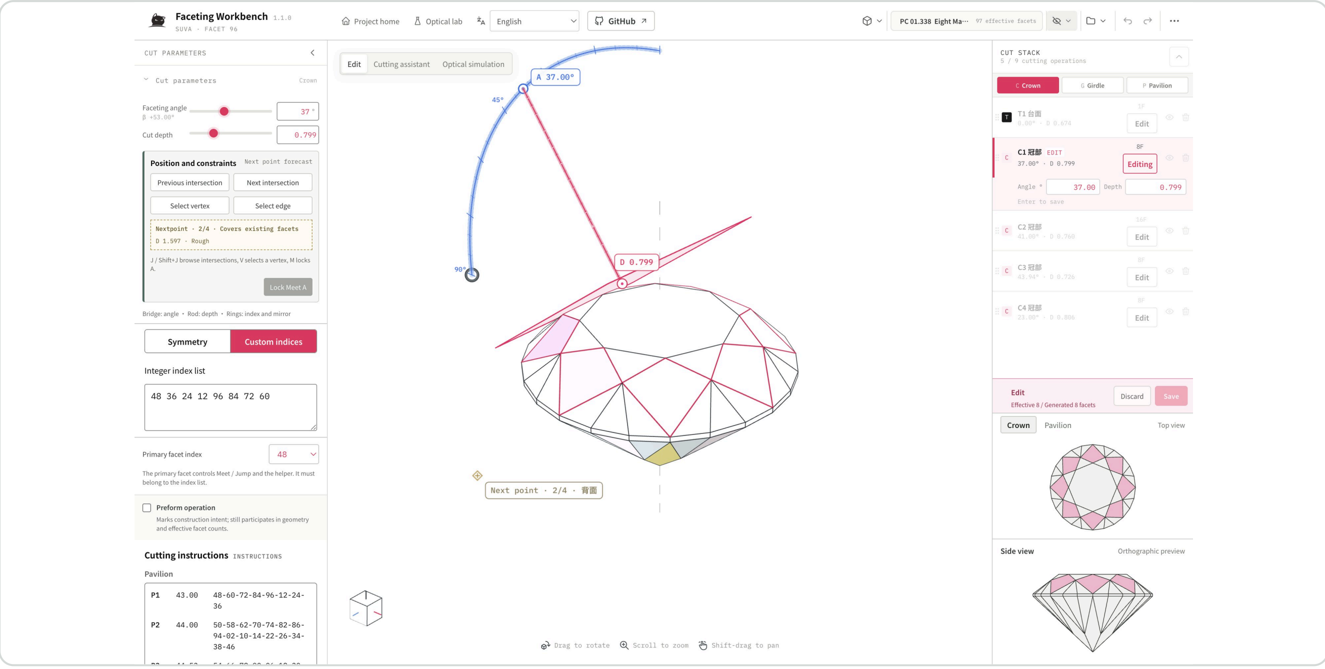
Before you continue

Committed cuts and material save automatically. Cards exclude unsaved previews; reloading does not restore drafts or old undo history. The card's upper-left delete control requires confirmation and cannot be undone. Export important designs as JSON.

PREPARE · EXERCISE FILES

Choose a starting point that reveals the question

Use Eight Main Highlight from the preset library for finished-cut observation. Cutting and Meet exercises use a simplified round built in stages, making new intersections easy to see. Exercises are not recommended finished cuts or production recipes.



Finished-cut example: PC 01.338 Eight Main Highlight, 97 effective facets, designed by Long, R H & Steele, N W. Open File → Browse cut presets and search its name.

Open the exercise

Create a project, then choose File → Import JSON and open 01-round-start.json. Rename it, for example "Round_edge_study_01", and export a baseline JSON. Import replaces the current design and can be undone; an unsaved preview requires confirmation first.

Write one design goal

For example: "Keep the round outline and add a light decorative rhythm to the crown." Judge each later change against this sentence.

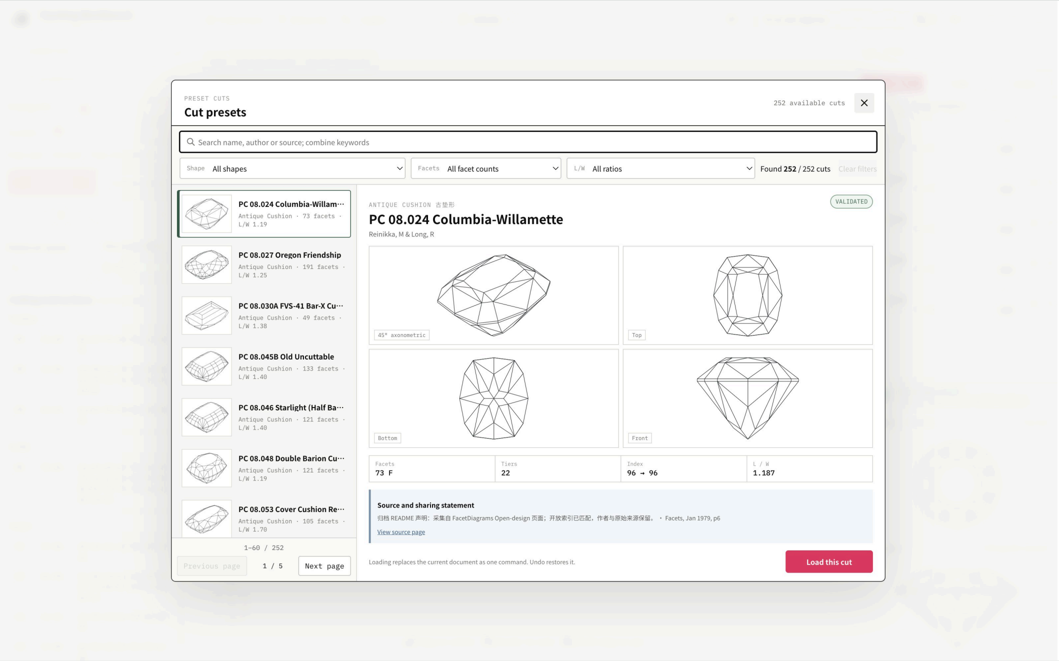
Exercise and comparison files

02 / 03 compare two ratios on the same edge; 04 uses two points; 05 lowers the crown; 06 adds four-direction decoration; 07 needs repair. File numbers identify alternatives, not a required cutting sequence.

START · CHOOSE A CUT

Start with an outline in the preset library

If you already want a cushion, oval or emerald shape, find a similar outline in the library. Treat the preset as a study: retain its author and source, and save your own variation separately.



Actual preset library: compare all four views to avoid perspective distortion.

1. **Open candidates**
   File → Browse cut presets. Combine shape, effective facet count and L/W filters. Search multiple words across name, author and source; each page holds 60 entries.

2. **Look from four directions**
   Top shows outline and table; front shows crown height and pavilion depth; bottom shows pavilion structure; axonometric shows the whole shape.

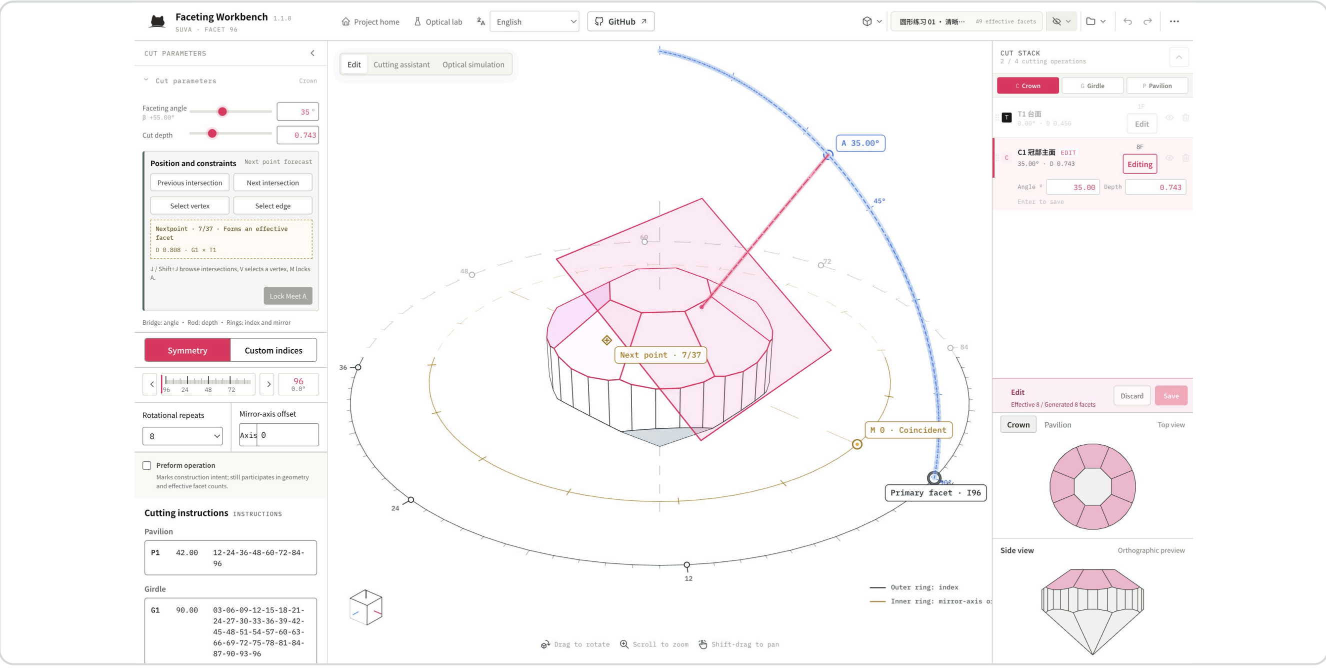
3. **Load and name**
   Load the chosen cut and rename it at the top. Export a baseline JSON first. Loading can be undone in one step; filtering and paging leave the design unchanged.

> **Pause and make a design judgment**
> Does the outline suit the intended setting? Does the table size suit the shape’s character? Describe any remaining concern as one concrete change.

START · WORKSPACE MAP

Find your next action in the workbench

Inspect the 3D shape in the center, adjust the current cut on the left, and review steps and two live views on the right. Each CUT STACK row is a group of facets edited together.



After clicking Edit on C1: the parameter panel, primary cutting plane and pink editing bar appear together.

Left: adjust the current facets

Faceting angle controls tilt, depth controls advance, and index controls direction. Repeats and mirrors arrange facets around the stone.

Upper right: steps and saving

Use C Crown, G Girdle and P Pavilion to filter the list; scroll for more cuts. Click Edit on a row. The green New control at the bottom becomes Save / Discard during editing.

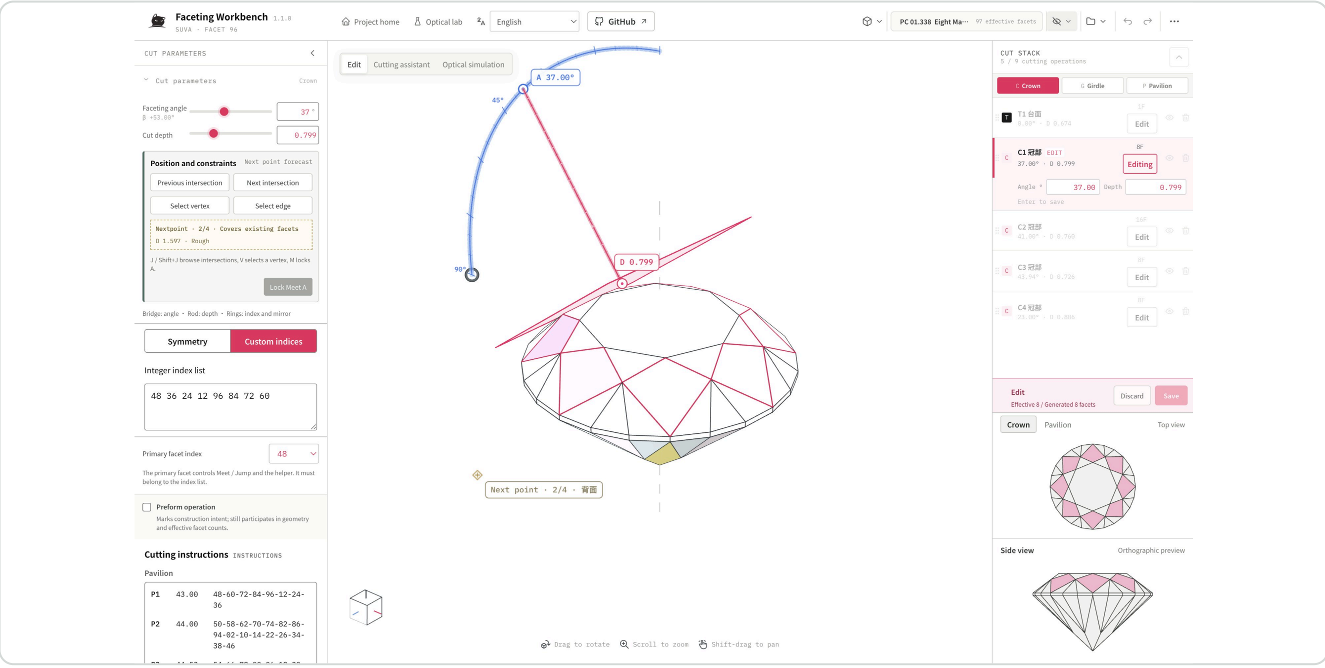
Top and canvas: inspect and manage

The top bar provides project home, optical lab, views, display, files and undo. The canvas mode switch selects Edit, Cutting assistant or Optical simulation. Drag to rotate, scroll to zoom, Shift-drag to pan. GitHub remains available in every mode and opens in a new tab.

OBSERVE · LIVE SIDE PREVIEWS

Check crown, pavilion and profile as you work

Judge facet rhythm and overall height together. Two orthographic windows follow the current shape while keeping fixed viewing directions as you rotate the central 3D view.



The upper window switches between crown/top and pavilion/bottom; the lower window stays in side view. Helpers are hidden so the actual intersections remain clear.

1. **Fix a starting point**
   Load Eight Main Highlight and edit C1 Crown. Adjust depth slightly and watch the central and side views update together.

2. **Compare facet groups**
   Choose Crown in the upper window to inspect facets around the table, or Pavilion for bottom intersections. This affects only that window, without rotating the central stone or changing cuts.

3. **Compare proportions**
   Use the lower window to compare crown height, girdle and pavilion depth. Check table position, outline continuity and new small facets. Save in CUT STACK if satisfied; otherwise discard.

Pause and make a design judgment

At the same viewing angle, does the added facet group make the rhythm clearer? Does the profile still match your design goal?

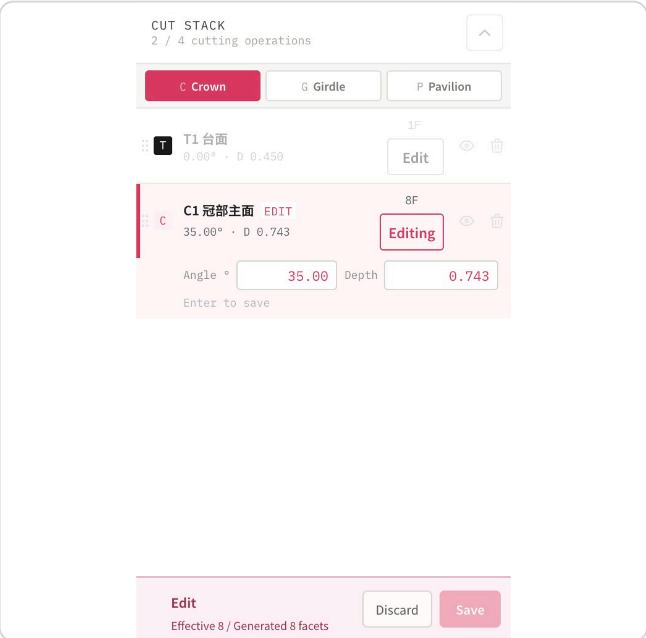
Before you continue

Side windows are live, read-only views: pale blue marks a new preview and pale pink the edited tier. Unsaved shapes can be compared but are excluded from projects and exports.

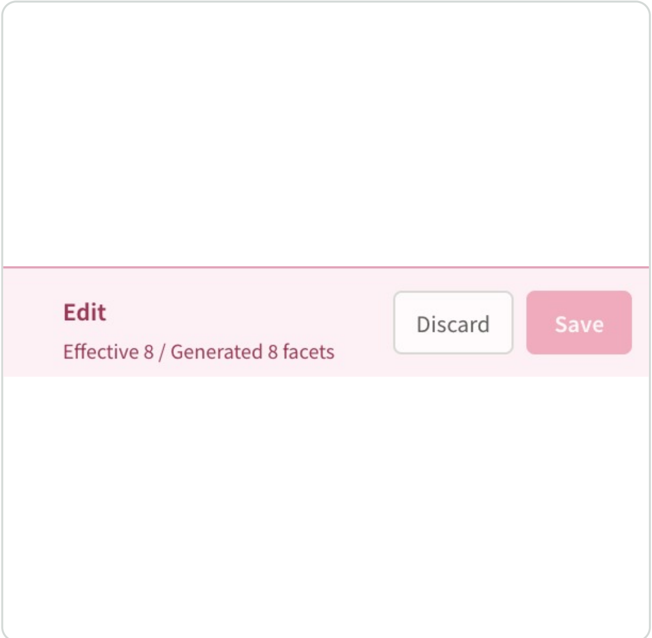
BASICS · EDIT AND SAVE

To change a facet group, click Edit

Every editable tier has a visible Edit control. The selected row says Editing. Clicking the name still renames the tier.



Tier list · Editing C1



Bottom of list · Discard or save

C1 is being edited. Save at the bottom of CUT STACK, not in the left parameter panel.

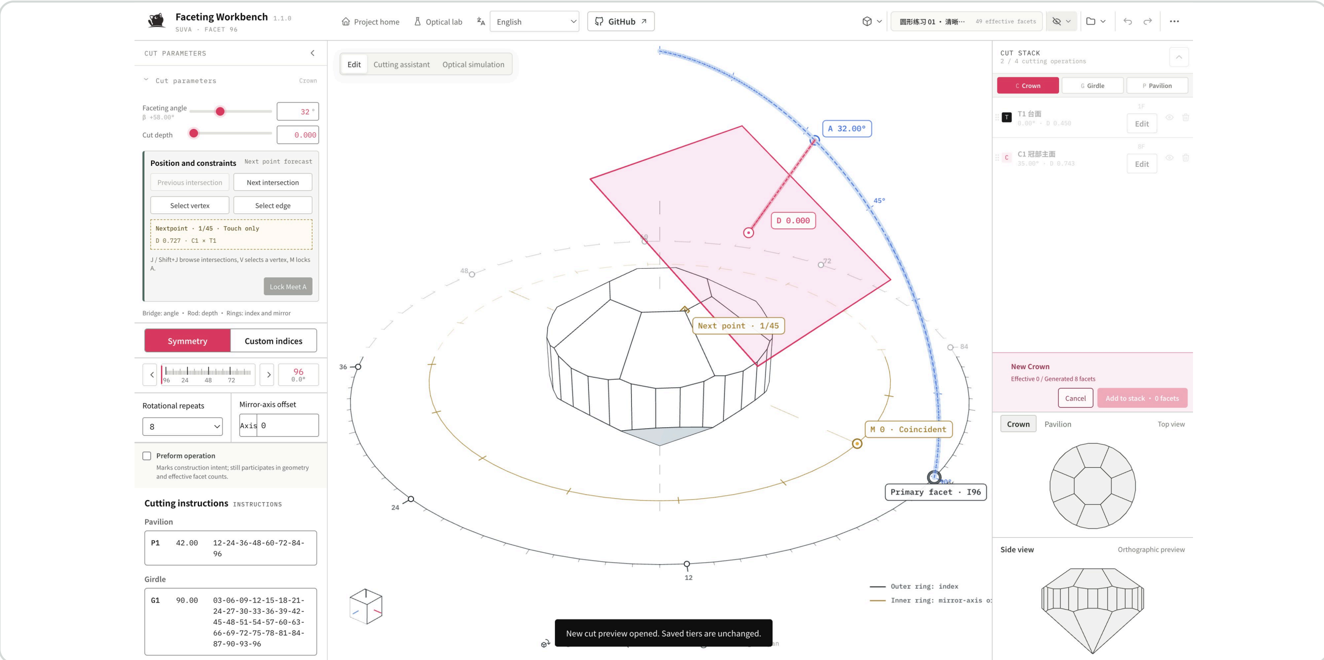
1. **Enter**
   Find C1 Crown mains in the C Crown list and click Edit. The initial view matches the saved shape.
2. **Try**
   Increase the angle slightly and use front view to inspect the tilt. Pink indicates the current edit, not the final material.
3. **Decide**
   Save the change if it works, or discard to retain the original. Saving returns to idle; Undo restores the previous step.

> **Pause and make a design judgment**
> Can you immediately tell which facets you are editing? Does the change support your stated goal? Being able to click Save is not the design outcome.

BASICS · WHAT THE PARAMETERS DO

Explore angle, depth and direction in small steps

Change one relationship at a time so you can identify its effect. Look at geometry before material. Blue new-cut previews and pink edit highlights are interaction feedback only.



The helper and primary plane show the direction of a new CUT. Actual facets are determined by intersections with the solid.

Angle: change the tilt

Crown and pavilion angles adjust in 0.01° steps. The table stays at 0°, the girdle at 90°. Single Meet allows angle changes; dual Meet determines the angle from two points.

Depth: decide how far to advance

Advancing the cutting plane along its normal changes intersections with neighboring facets. Enter exact values when needed. The handle endpoint is not a gemstone vertex.

Index: turn the facet group

There are 96 index positions per revolution; 96 and 0 indicate the same direction. The helper follows the selected primary facet.

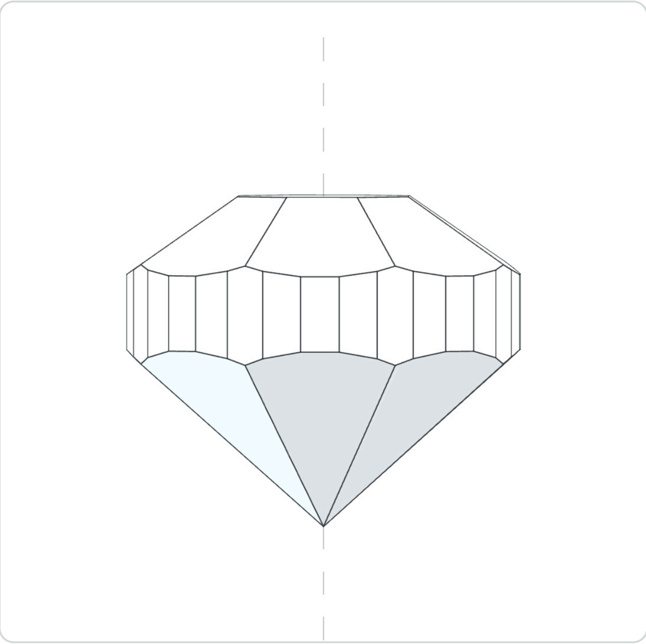
Why saving may be unavailable

A new cut must leave at least one effective facet. No effective facet, stale references or unsatisfied constraints produce an explanation. Understand the reason before adjusting, releasing or canceling.

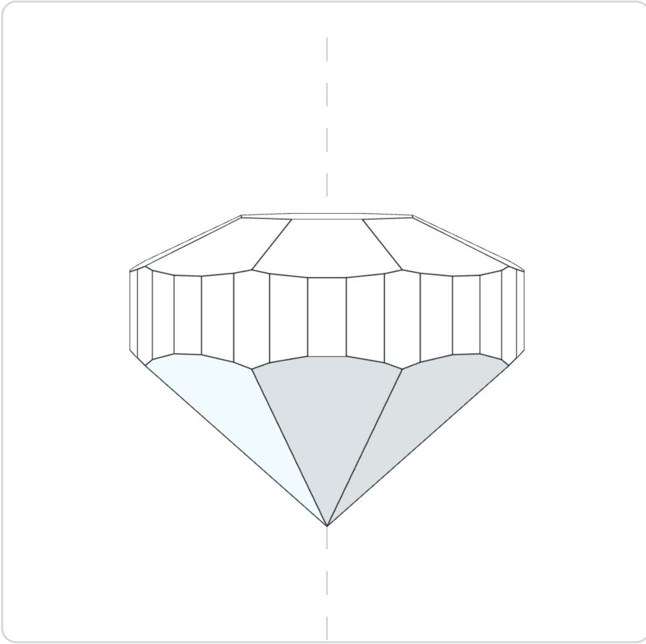
EXAMPLE A · OVERALL PROPORTIONS

Compare a tall crown with a lower crown

Goal: keep the round outline and see whether a lower crown suits your setting height or style. The group transform includes the table, helping retain continuity between tiers.



Original · Height 100%



Lower crown · Height 70%

Compare from the same front view. Only crown height changes; the pavilion stays unchanged.

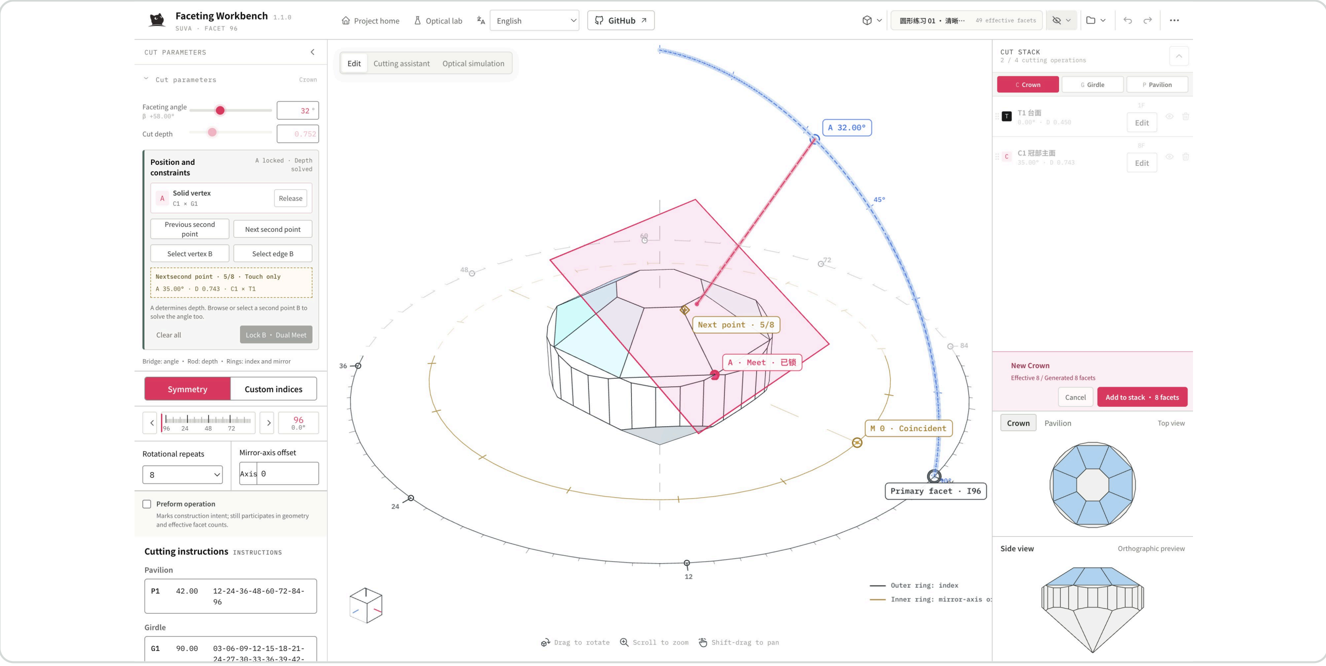
1. **Open the original**
   Import 01-round-start.json and click Group transform in the crown list.
2. **Try 70%**
   Set height scale to 70%, leaving translation and rotation at 0. Inspect front and side views, then save. Import 05-low-crown.json to compare the result.
3. **Judge, rather than copy**
   Check the table-to-crown transition and whether the height suits your design. 70% is a comparison condition, not a universal optimum.

> **Before you continue**
> Group transforms also combine vertical translation and rotation by whole index teeth. The crown includes the table; the pavilion is separate. Existing Meets may need review after a source tier changes.

EXAMPLE B · JUMP AND SINGLE MEET

Pass the primary plane through an existing intersection

Goal: meet an existing structure precisely. Jump browses candidate positions; Meet remembers a chosen position as a constraint.



After locking A, the target point determines depth; faceting angle remains adjustable.

1. **Find a position**
   Create a crown cut and use Next / Previous intersection, or Select vertex and pick a point on the solid. Check that the highlighted sources are the intended intersection.

2. **Lock A**
   Click Lock Meet A after inspecting the preview. When you change angle, depth adjusts to keep the primary plane passing through A.

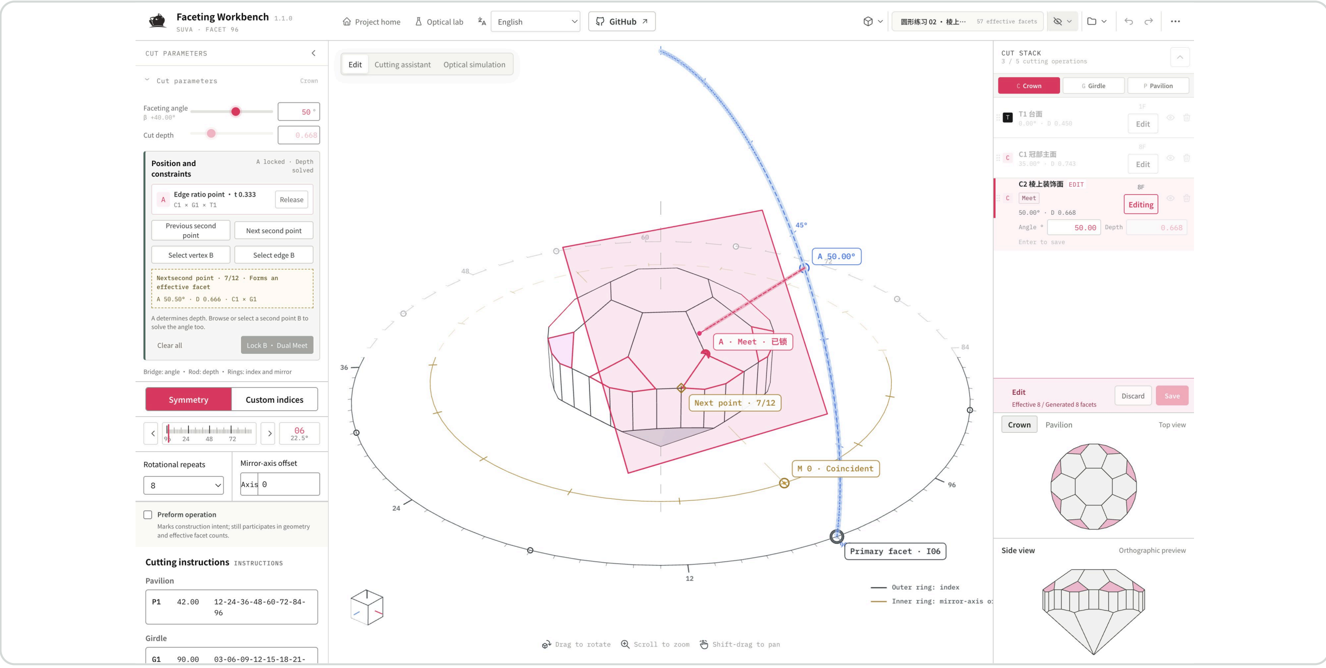
3. **Check the whole ring**
   Repeated facets cut together, but the constraint applies to the primary facet. Rotate the stone and check other directions before adding the cut.

> **Before you continue**
> This example adds a single-point crown cut to the round exercise. Later edge examples use the same starting point. Girdle and fixed table cuts do not support Meet / Jump.

EXAMPLE C · EDGE RATIO

Choose where decoration starts on an existing edge

Goal: add small facets between eight crown mains, turning sharp ridges into a decorative band of varying width. An edge ratio gives you a repeatable position to compare.



Variant 02: primary I6, 8 repeats, 50°, passing through one third of the sloping edge between C1's I96 and I12 facets.

1. **Identify the edge**
   Import 02-edge-third.json and edit C2. The A source description and canvas marker identify the sloping edge, not a horizontal table edge.

2. **Try from the beginning**
   Import 01, add a crown cut, and set index 6, repeat 8 and angle 50°. Select the sloping edge, set its ratio to 1/3, then lock A.

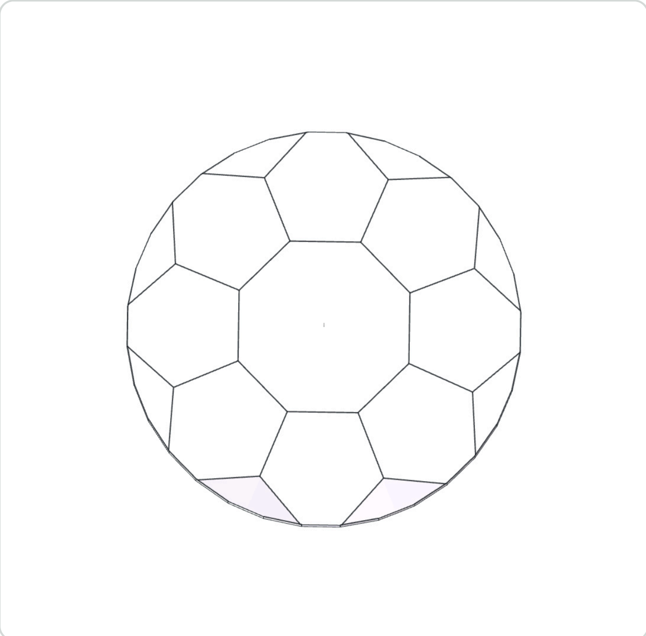
3. **Check the target position**
   Here the stable edge direction runs from girdle toward table, so 1/3 is nearer the girdle. For another edge, temporarily try t = 0 and 1 to identify its ends. Do not assume every edge runs upward.

> **Before you continue**
>
> An edge Meet passes the plane through one point on an edge; it does not place the whole edge in the plane. If you chose the wrong edge, select the correct source instead of compensating with depth.

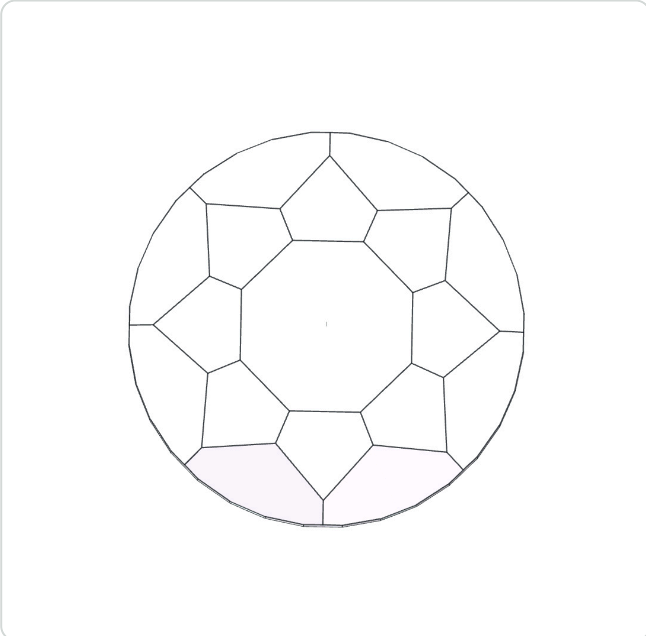
EXAMPLE C · COMPARE SHAPES

One edge, two ratios, two rhythms

Keep 50°, I6 and 8 repeats unchanged; move A from 1/3 to 2/3 of the same edge. This isolates how much crown area the decoration occupies.



1/3 · Variant 02



2/3 · Variant 03

Both files retain 57 effective facets. Equal facet counts do not imply equal facet widths or equal space around the table.

> **Workflow**
> Import 02, edit C2, release A, then select the same sloping edge at 2/3. Lock Meet A and save. You can also import 03 directly. A locked Meet is not adjusted by freely dragging depth.

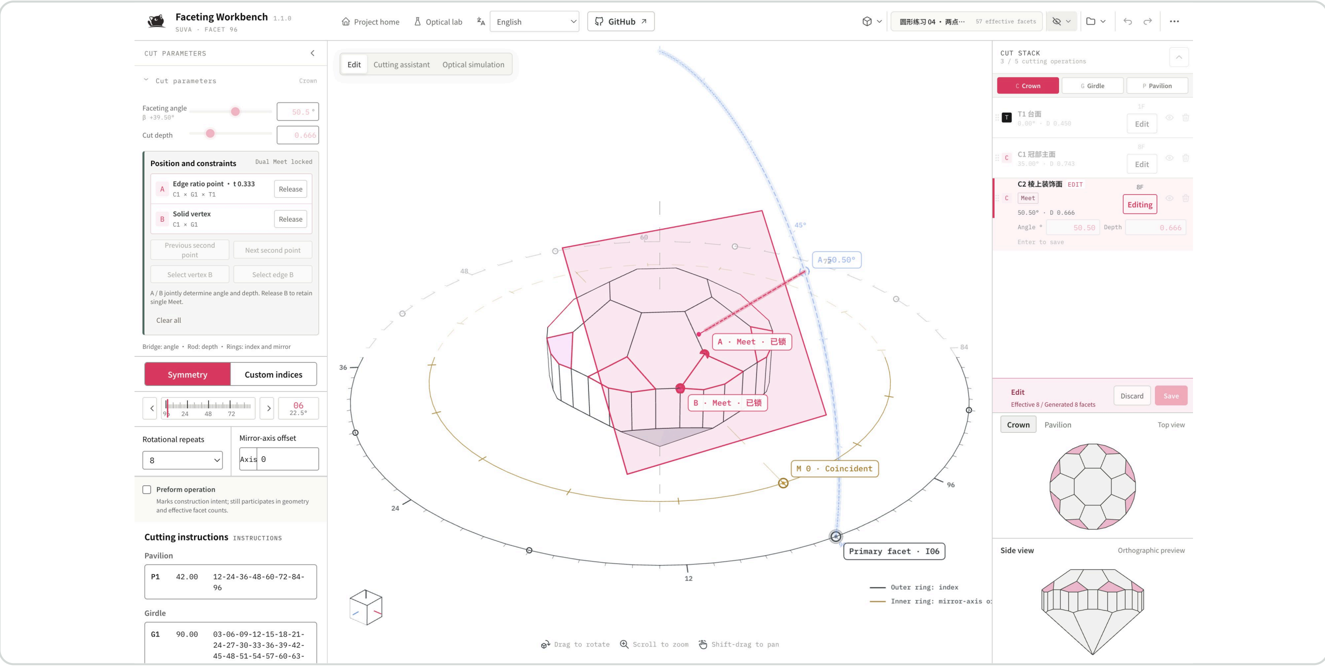
> **Observe three relationships**
> Look at the new facet tips near the table, the remaining crown-main area and the eightfold rhythm in top view. Check axonometric view to rule out a viewpoint illusion.

> **Let the designer decide**
> Do you want broader crown mains or more subdivision? Write a reason to use each alternative before choosing. This manual does not declare one more beautiful.

EXAMPLE D · DUAL MEET

Use a second point to determine tilt

Goal: satisfy two meeting relationships at once. With a fixed primary index, two distinct points can determine the plane's tilt and position together.



Variant 04: A is at one third of the sloping edge, B an existing vertex. The solved angle is about 50.50°; depth is solved with it.

1. **Inspect A and B**
   Import 04-dual-meet.json and edit C2. Inspect both markers and sources to understand which intersections they preserve.

2. **Try a second point**
   Starting with A locked, browse Next / Previous second point or select a second vertex or edge point. Inspect the candidate, then click Lock B · Dual Meet.

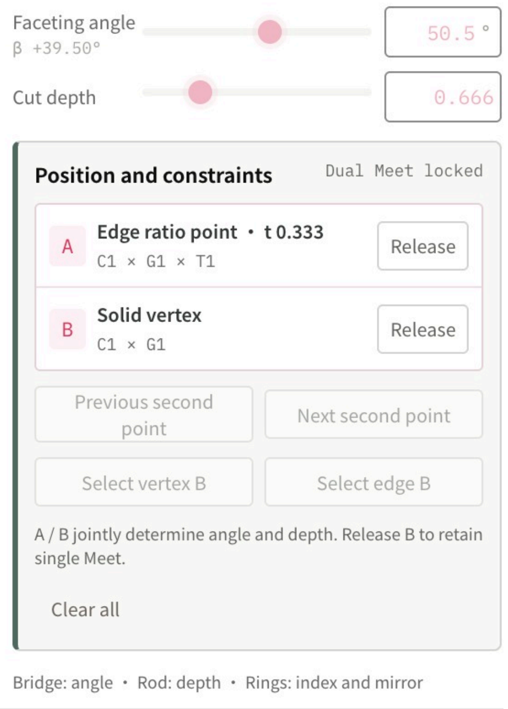
3. **Inspect the result**
   Angle and depth becoming read-only is expected. Use top and axonometric views to judge whether meeting both points is worth changing your preferred tilt.

> **Before you continue**
>
> A suitable second-point solution may not exist. Selecting two points does not make a design better by itself; dual Meet is a precision construction tool.

MEET · CANCEL AND RELEASE

Step back while keeping the choice you made

When exploring, know how to return. Distinguish browsing a second point from having both points locked.



A and B are shown separately so either source can be released. Locked parameters are visibly disabled.

B is still a preview

Canceling the second-point preview restores the original single-A angle and depth. You do not need to remember or re-enter them.

B is locked

Release B to retain A and adjust angle again. Release A to promote B to A. Clear all to restore free angle and depth.

When no solution is available

If points coincide, the direction is unsuitable or the target is out of reach, change the point or primary index. The system does not silently choose another primary facet.

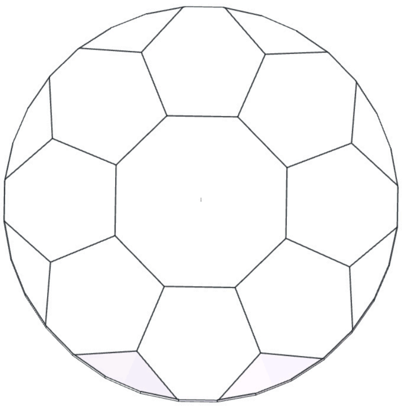
Escape order

Close the floating panel or current picking / ratio tool first, then leave the second-point preview, then cancel the CUT. Watch the state rather than pressing repeatedly.

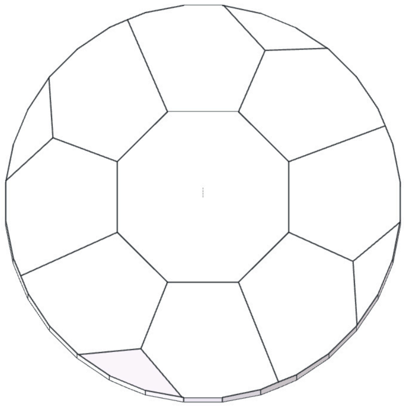
EXAMPLE E · CUSTOM RHYTHM

From eight repeats to four main directions

Goal: concentrate decoration in four directions to establish hierarchy. Custom indices specify where facets appear; the primary facet specifies the direction used for positioning.



Eight directions · 57 facets



Four directions · 53 facets

Variant 06 retains only I6, 30, 54 and 78. Removing four decorative facets changes the open areas; it does not remove all crown facets.

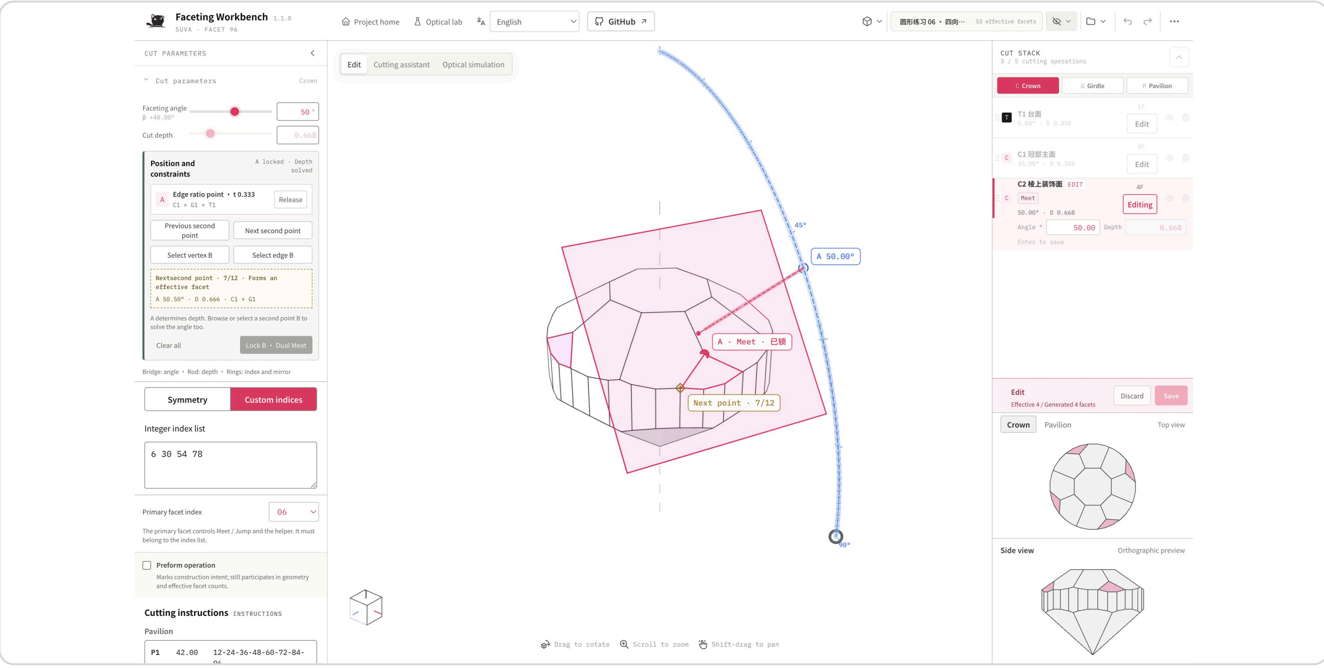
1. **Specify the positions**
   Import 02-edge-third.json, edit C2, switch to Custom indices and enter 6, 30, 54, 78.
2. **Choose the primary facet**
   Select 6 as the Primary facet index, then check the A Meet. The primary facet must belong to the set; it cannot be removed while a constraint uses it.
3. **Compare the composition**
   Use top view to check whether the four directions suit your intended setting. Recheck the solid after rotating or changing the index set.

> **Before you continue**
>
> This example still has fourfold symmetry. Asymmetric sets are supported, but require facet-by-facet inspection; symmetry is not added automatically.

ORGANIZE · BEYOND FACET COUNT

Express intent with repeats, mirrors and preforms

A facet group may create rhythm or provide a source for a later cut. Meaningful step names make it easier to resume than remembering long parameter lists.



The helper and Meet / Jump follow the selected primary facet.

Repeat: one rule around the stone

8 repeats generate eight explicit planes. Later cuts may cover some earlier facets; the original operations remain available for undo and redo.

Mirror: add a second set of directions

Adjusting mirror-axis offset adds mirrored members; coincident planes are deduplicated. Observe the alternating rhythm in top view instead of maximizing facet count.

Preform: describe this step's purpose

Mark a normal crown or pavilion tier as a Preform operation when it prepares for later steps. This marker does not hide, delete or disable the cut.

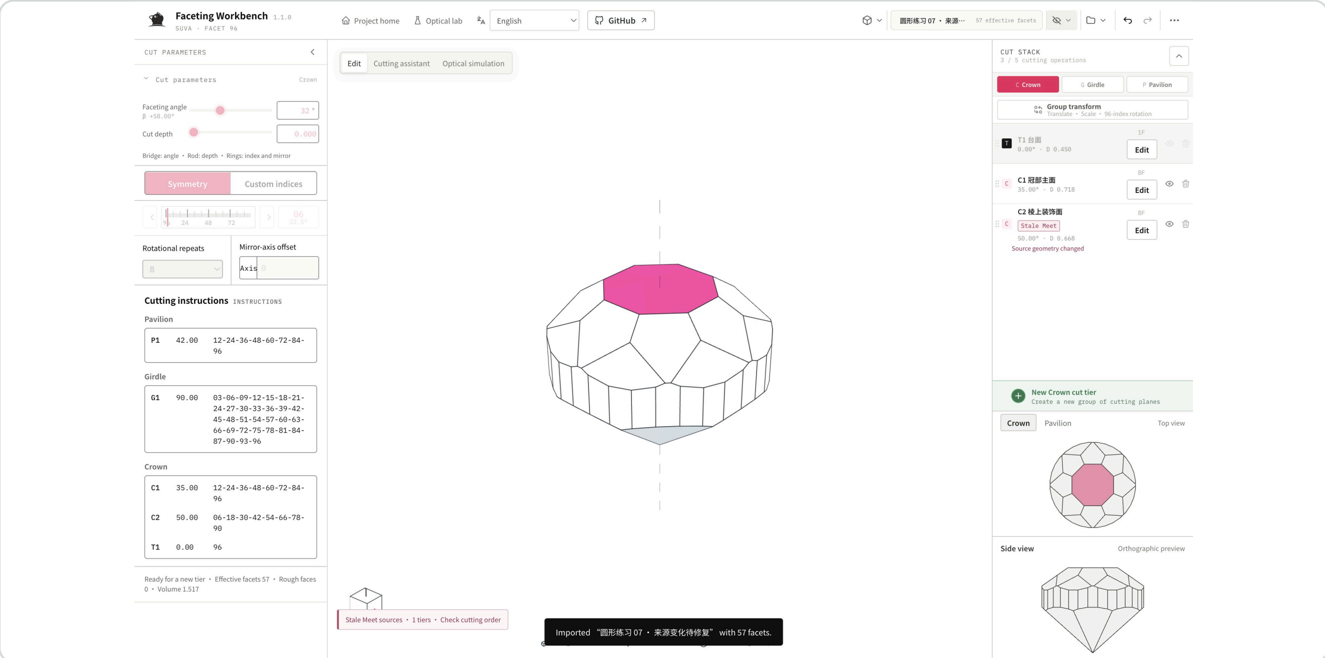
Name it for your next collaborator

Click a tier name to rename it, for example "Crown mains" or "Four-way decoration". Toggle visibility to inspect effects; drag non-table tiers to reorder. These changes may invalidate later Meet sources.

EXAMPLE F · REPAIR REFERENCES

After changing an earlier shape, what happens to Meets?

Goal: retain saved cutting planes while identifying meeting relationships that need a new design decision. A stale source does not automatically recut the stone; the warning asks you to reconsider intent.



Variant 07: raising C1 Crown mains by 0.03 changes the source of C2’s original edge point.

1. **Read the warning**
   Import 07-source-changed.json and inspect C2’s stale marker. Compare it with 02: which intersections no longer match your intent?

2. **Select again**
   Edit C2, release the stale source, and select the required edge point or vertex on the solid before this cut. To retain the original intent, try one third of the same sloping edge.

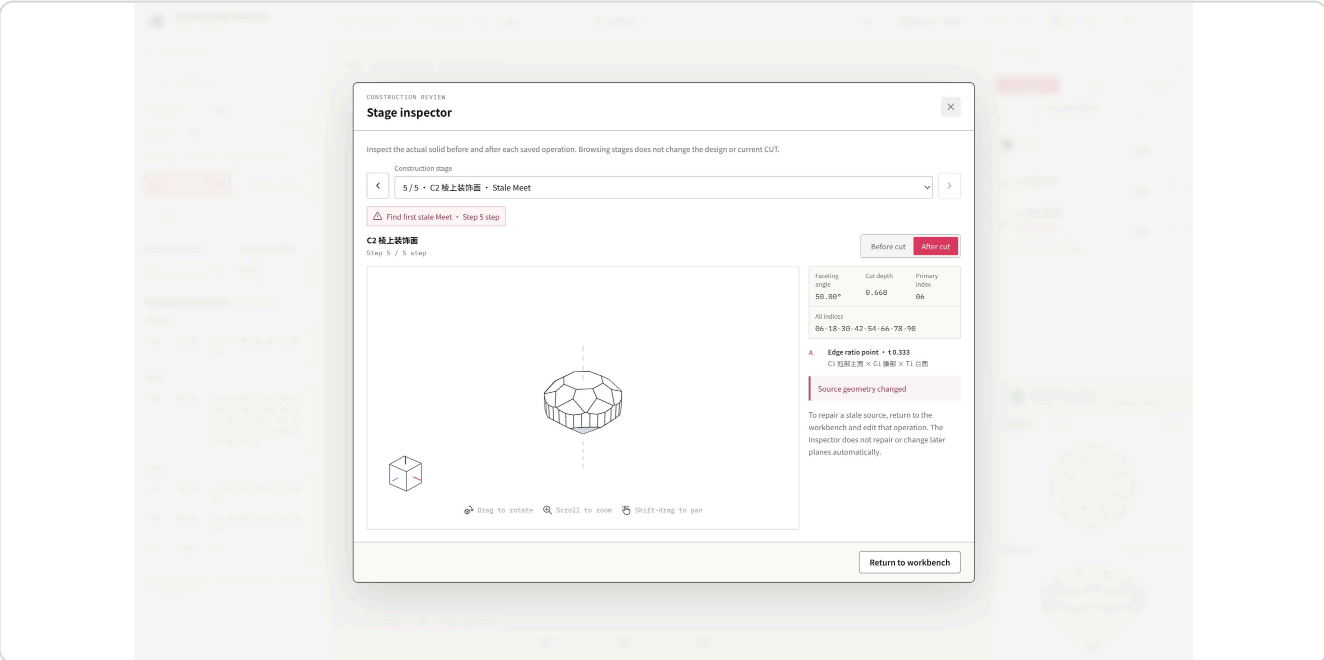
3. **Decide again**
   Inspect and save the new preview. If the Meet is no longer needed, clear the constraint and retain a free construction. Do not choose an arbitrary point just to remove the warning.

> **Before you continue**
>
> Hiding, deleting, reordering or editing a source can make it stale. Saved planes remain intact and a warning appears; downstream cuts are not all changed automatically.

DIAGNOSE · STAGE INSPECTOR

Inspect stages to understand how the shape changed

Goal: break the shape into understandable design steps. Use the inspector to review mains, decoration and source relationships, alone or with another designer.



The inspector pairs stage geometry with parameters and sources. This example focuses on C2's stale reference.

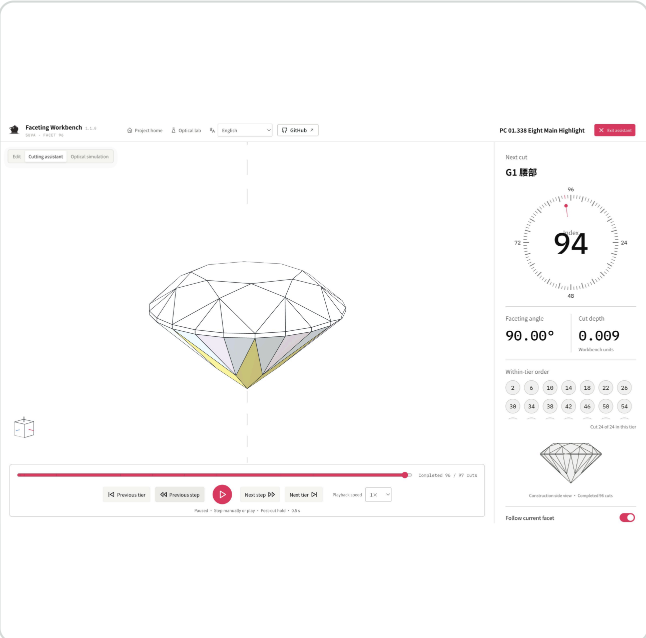
1. **Open**
   More tools → Stage inspector. Select an operation or locate the first stale Meet.
2. **Compare before and after**
   Switch Before cut / After cut to see what the step actually leaves. A hidden operation produces no difference.
3. **Return with a question**
   The inspector is read-only and preserves your editing workspace. Close it and edit the relevant tier to repair the design.

> **Pause and make a design judgment**
> Does this step add clear structure or unnecessary fragments? Does it still serve as a preform? Keeping or reverting it are both valid choices.

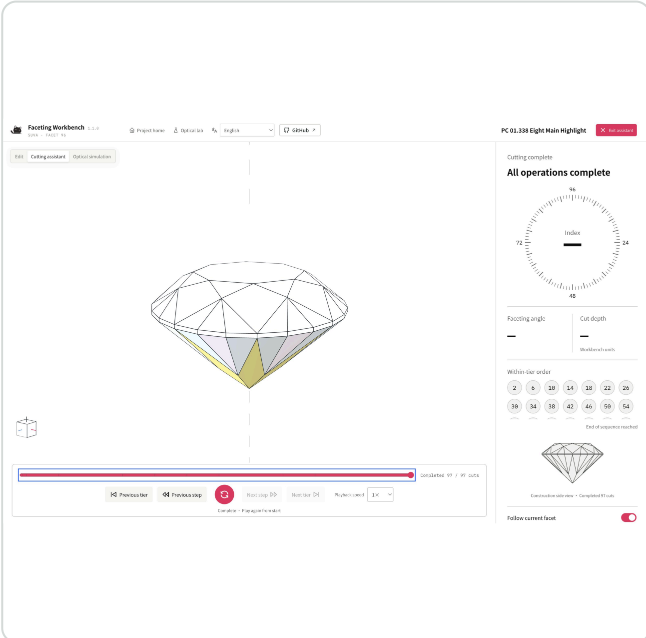
DEMONSTRATE · CUTTING ASSISTANT

Replay the actual cutting order before delivery

Check the intended cutting order or show a collaborator how the stone is built. The assistant replays the committed design from the rough, one cut at a time, in CUT STACK order.



86/97 cuts completed · Next cut and intermediate shape



Step 97/97 · Complete

Actual Eight Main Highlight replay: pink marks the next plane, while the intermediate solid shows completed cuts. The sequence uses committed CUT STACK only, excluding drafts.

1. **Enter**

   Canvas mode switch → Cutting assistant. Start at step 0 with the rough. Hidden tiers do not cut; the fixed table is always one step.

2. **Watch by cut or by tier**

   Use Next / Previous step on the bottom bar for individual cuts, or Next / Previous tier to jump groups. Play runs at 0.5× / 1× / 2×. Drag the timeline to seek; manual steps and backgrounding pause playback. The right panel shows the next index, angle, depth and within-tier order. Completed counts cuts already made.

3. **Return to editing**

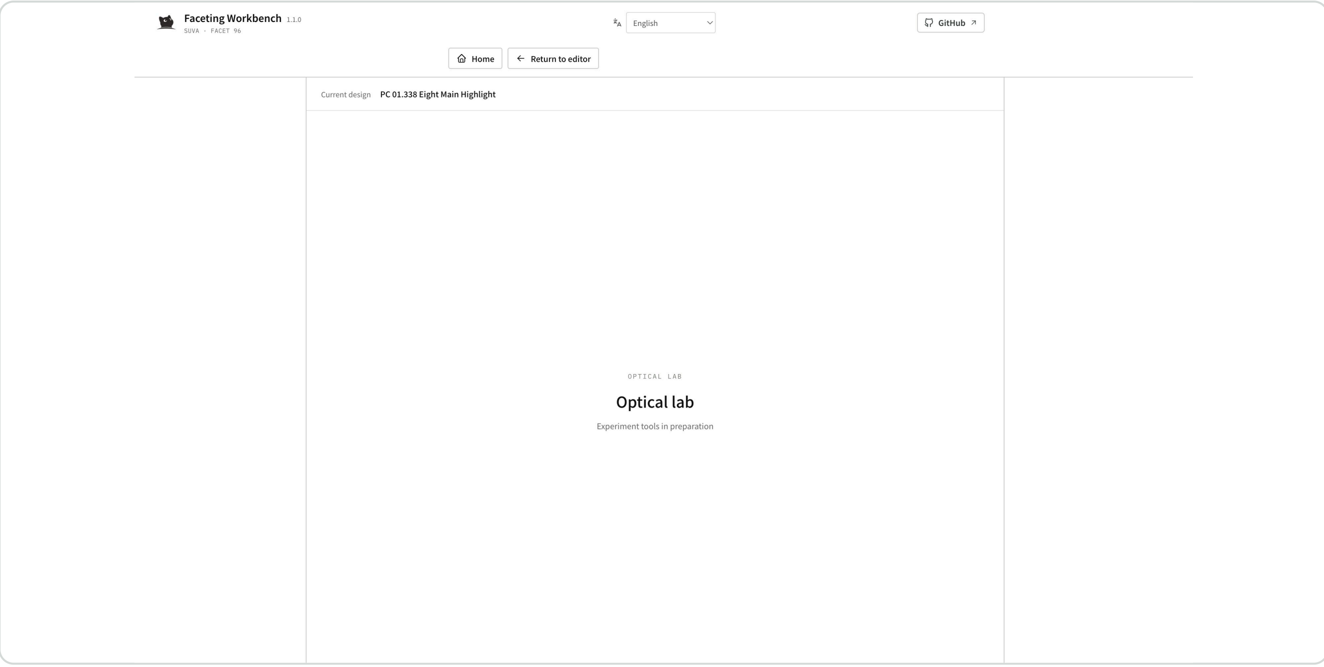
   Click the pink Exit assistant button or press Escape to restore the previous workspace. The assistant is read-only and adds no history. After each cut, it holds 0.5 seconds, then eases to a 45° offset view; the next plane appears once settled. Adjust transition speed on the right. Dragging pauses following; the switch resumes it. Use Stage inspector for stale-source diagnosis.

> **Pause and make a design judgment**
>
> Which facet group's ordering needs explanation to the faceter? Does the sequence express your intent, or should you revise CUT STACK?

NAVIGATE · OPTICAL LAB

A space for experiments on the current cut

Enter the optical lab from home or the editor. It retains the current project and the workbench's navigation and side-panel structure for future experiments on the same stone.



The optical lab currently shows the project name and return controls. Experiment tools are in preparation.

1. **Enter with the current design**
   Click Optical lab in the editor and verify the current design name. From home it uses the project opened in this session, or states that none is selected.

2. **Return to your workspace**
   Return to editor resumes the project; Home opens the project list. Changing pages neither commits nor cancels a CUT and does not treat previews as saved content.

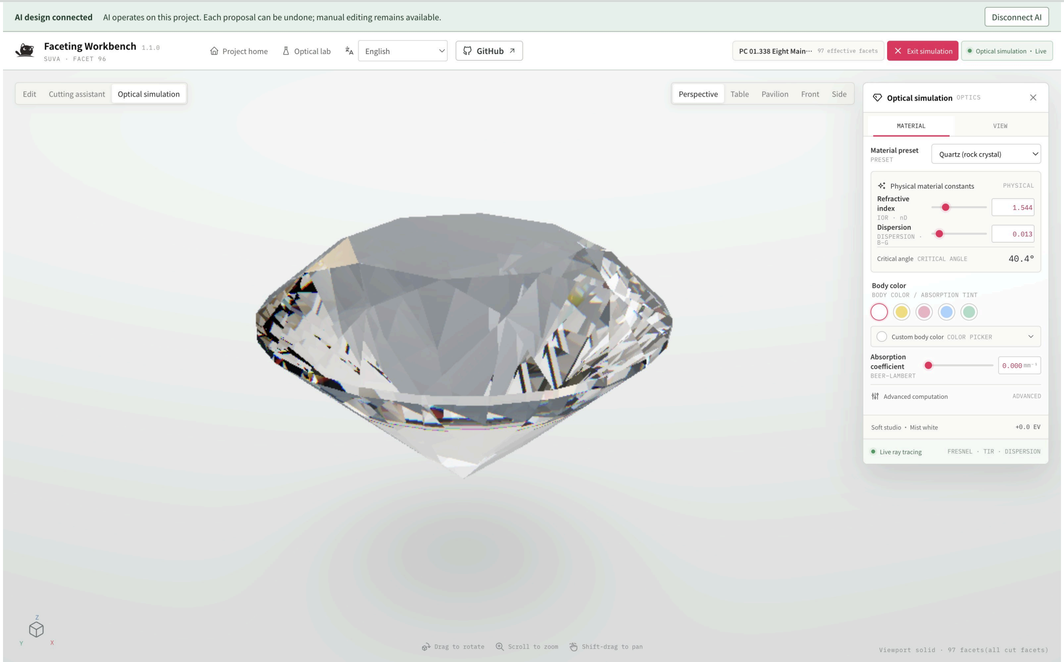
3. **Observe material now**
   Return to the editor and choose Optical simulation in the canvas mode switch or display menu. The next two pages explain comparison under fixed conditions.

> **Before you continue**
> This page demonstrates context and navigation only. No experiment results, scores or new simulation tools are provided. Use Optical simulation for material observation in this version.

OPTICS · OBSERVE DIFFERENCES

Fix viewing conditions before comparing materials

Once the geometry is clear, enter Optical simulation from the canvas mode switch or display menu. Editing tools hide so you can focus on material, light and outline. View changes ease to a stop; dragging takes control. Exiting restores the editing workspace.



Use the upper-right bar to switch views. This actual quartz-material simulation of Eight Main Highlight shows one viewing condition, not a prediction of physical appearance.

Keep conditions consistent

Use the same material, body color, studio, exposure and viewpoint when comparing two cuts, so differences remain attributable.

Inspect the structure

Observe light distribution around the table, whether mains remain legible, and whether changes during rotation match your intended rhythm.

Save each alternative

Material is saved with the document. Keep separate JSON files for A/B alternatives and record viewing conditions. Exiting simulation does not commit unsaved cuts.

Before you continue

Simulation supports design comparison; it does not guarantee physical fire, brightness scores or yield. Material, size, cutting accuracy and viewing environment also affect the real result.

OPTICS AND GEOMETRY · DESIGN JUDGMENT

Four questions for a useful design review

Do not choose only the prettiest perspective image. Examine both alternatives in the same order and explain why you keep or reject each.

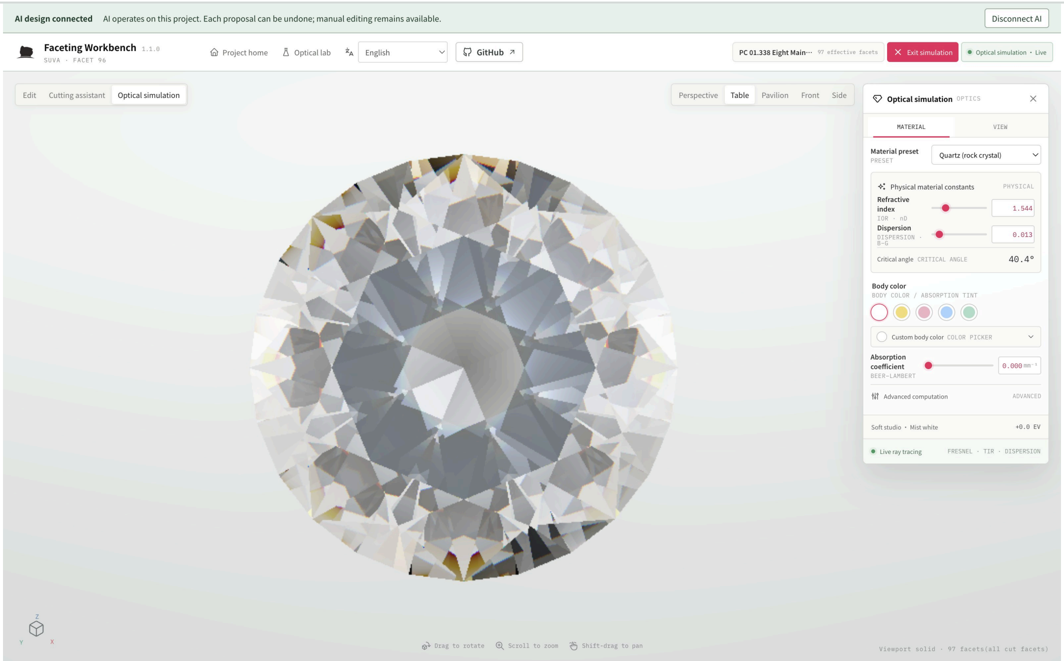


Table view makes outlines and facet groups easier to compare. Rotate slightly to understand changes in light distribution.

Top: does the rhythm work?

Do main and decorative facet proportions support the goal? Is the space around the table intentional or overcrowded?

Front and side: do the proportions fit?

Does the combination of crown height, girdle and pavilion depth suit the intended form and setting direction?

Rotate: can you explain the changes?

With fixed material and exposure, rotate slightly and compare light structure. Avoid hiding large dark areas with overexposure or mistaking studio reflections for facet edges.

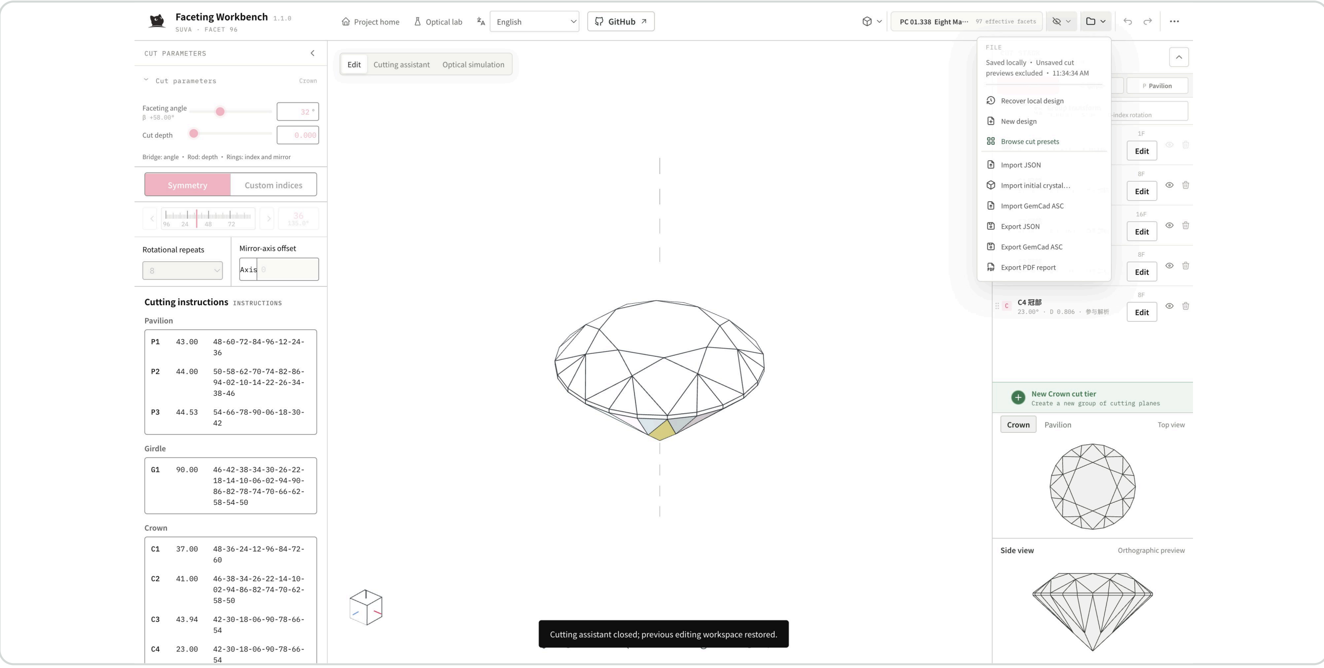
Return to geometry: is the shape clear?

Exit simulation and check intersections, missing facets and remaining rough. An optical image does not replace a geometry check.

FILES · SAVE AND DELIVER

Deliver the design and the ability to keep editing

Provide an editable master file, a report for discussion and a clear design note, alongside any presentation image.



Click outside a menu or press Esc to close it. The file menu offers JSON, ASC and PDF; Chinese tier names are supported. Save the CUT before exporting the intended design.

Continue designing: JSON

Preserves all tiers, Meet sources, preform intent and optical material. Use it as the master file; unsaved previews are excluded.

Read and discuss: PDF

The technical report contains geometry views, effective facet parameters and construction or stale-source notes. Choose whether to include the detailed girdle table. Check names and views before sending.

Industry exchange: ASC

Exchanges effective cutting planes and refractive index. Repeats and mirrors expand; Meet intent, preform purpose and construction history are not retained. Read preflight before confirming and keep a JSON copy.

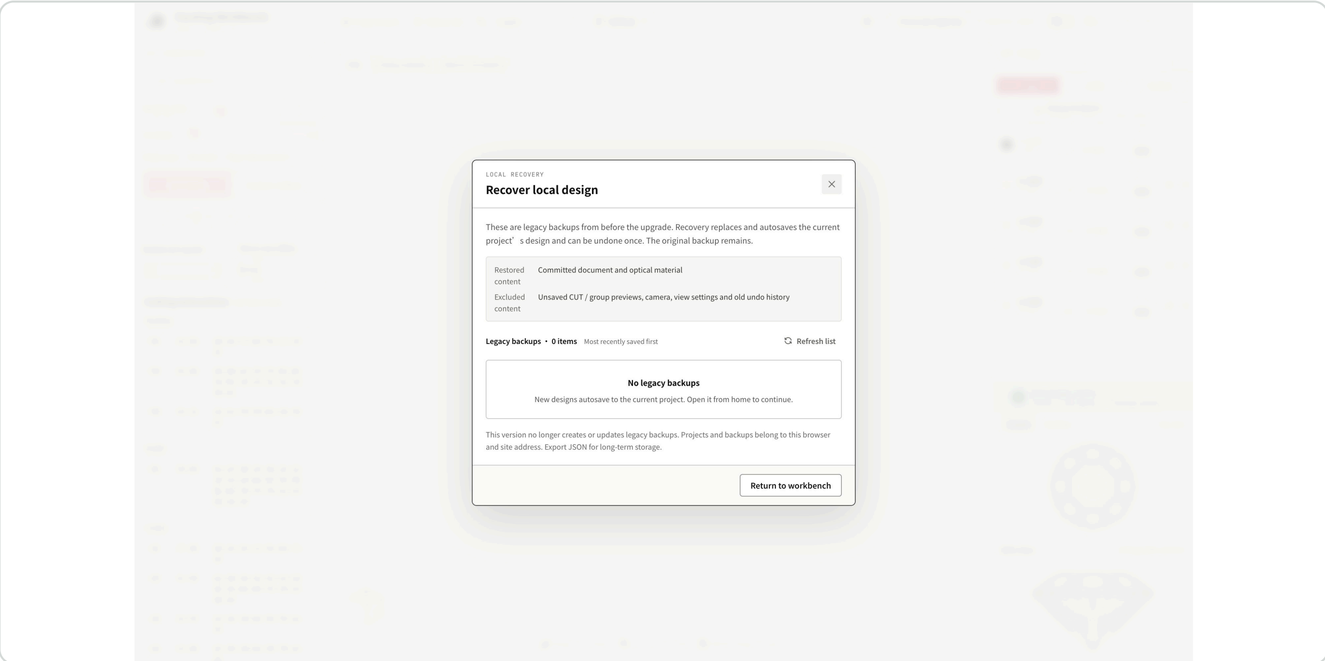
Before you continue

Before importing ASC, check the gear, 96-index mapping, scale and warnings. Incompatible mappings, preforms or table definitions block import. Do not silently alter files to bypass warnings.

PROJECTS · RECOVERY · SHORTCUTS

Save projects and recover older designs

Projects retain the latest committed document and material but do not replace exported JSON archives. Different site addresses or browsers have separate project storage.



File → Recover local design lists legacy backups only. If empty, continue an autosaved project from home.

Project saving and legacy backups

Committed cuts and material save to the current project. Legacy backups migrate once to home and are no longer updated. Manual recovery replaces and saves the current design, can be undone once, and leaves the backup intact. Drafts, camera and old history do not survive reload. Concurrent windows detect save conflicts and pause autosave; reload, save as a new project or export JSON to resolve them.

Useful shortcuts

Drag to rotate, Shift-drag to pan, scroll to zoom, 0 to reset. J / Shift+J browse candidates, M locks A, B locks the second point, V toggles vertex picking. Text editing takes priority in input fields.

When you need an explanation

More tools → Help center. Escape closes help or the active menu without changing the design. If saving fails, retry, export JSON or use Manage local projects to delete unneeded projects and free space. Archive before deleting projects or clearing browser data.

DESIGN REVIEW · DELIVERY CHECKLIST

Complete a design, not just a sequence of clicks

Review the work as a designer: can you discover the controls, understand changes, make trade-offs and hand over the result? Successful interactions are the foundation, not the whole outcome.

State the goal

Use one sentence, for example: "Add lighter four-way decoration while retaining eight crown mains."

Show before and after

Compare original and revised shapes side by side at the same angle. Identify the visible changes, beyond parameter screenshots.

Explain the choice

Describe your preferred outline, open areas or facet rhythm. Record remaining concerns and the next experiment.

Complete delivery

Ensure no CUT is unsaved; address stale sources; check multiple views; export the JSON master and PDF report, with ASC if needed.

Give actionable feedback

Describe "My goal → Where I clicked → What I saw → What I expected", with a screenshot and JSON. Unclear controls and confusing meanings are valid feedback too.

Pause and make a design judgment

Can you explain when group proportions, edge ratios, dual Meet and a custom primary facet helped with your own design problem?

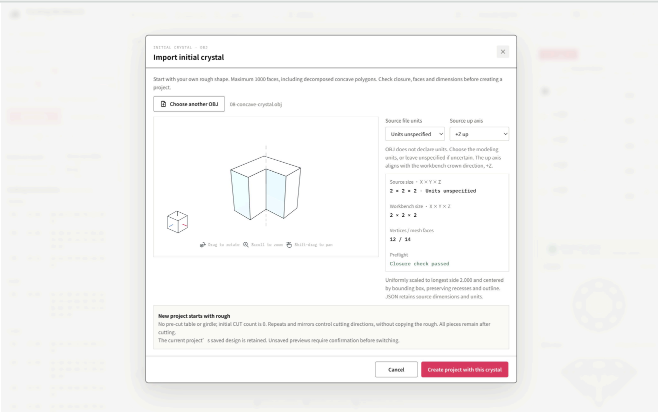
Before you continue

This manual and original exercises are provided by the faceting workbench. Preset authors, sources and third-party font / reference rights remain as documented. The applicable version is printed in the footer.

YOUR CRYSTAL · OBJ IMPORT

Keep the rough’s recesses, then plan the cuts

For a starting shape with recesses or an irregular outline, preserve its shape before deciding what to remove. Repeats and mirrors arrange cutting directions; they do not make asymmetric rough symmetric.



Actual preflight of exercise 08’s L-shaped crystal: inspect the recess, source dimensions and up axis before creating a separate project.

1. **Import a real starting shape**
   New project → Upload blank, or File → Import initial crystal. The 1,000-face limit includes decomposed faces; failed preflight explains why. Open docs/manual/examples/08-concave-crystal.obj: an original L-shaped exercise measuring 2 × 2 × 2 in unspecified units, +Z up.

2. **Inspect shape and proportion**
   Keep Units unspecified and +Z up. Rotate the preflight model and check the L-shaped notch. For your own OBJ, choose its original units and orientation. The longest side scales uniformly to 2.000 and the bounding box is centered.

3. **Start your cutting sequence**
   Click Create project with this crystal. It starts with no CUTs and no automatic table or girdle. Add a cut at the bottom of CUT STACK; start with one repeat and compare the original recess with the new facet.

4. **Keep a recoverable design**
   All pieces remain if a cut separates the solid. Clear cuts restores the original rough. JSON saves the rough and operations; crystal projects do not support ASC export, as preflight explains. Canceling import preserves the old project; switching asks about unsaved previews.

Pause and make a design judgment

Compare before and after in the same top and front views: did the new plane preserve the recess you wanted? Which repeated directions actually intersected the rough?

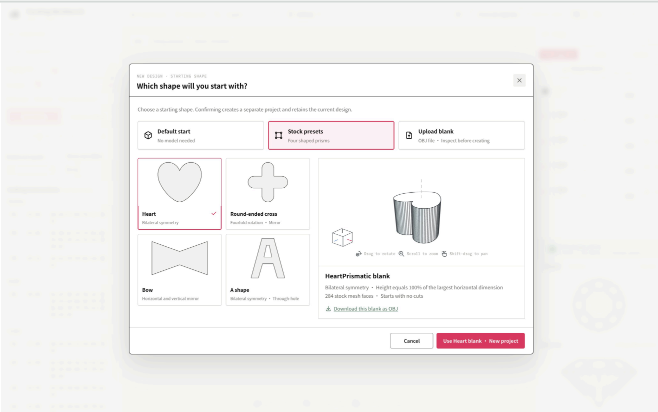
Before you continue

This exercise teaches the relationship between rough and cutting direction, not crystal measurement or a production recipe. If preflight fails, repair closure and face connectivity in a modeler, then re-export OBJ.

STOCK PRESETS · ORIGINAL OUTLINES

Design from a shaped blank

Start directly with a heart, round-ended cross, bow or an A shape with a through-hole, without preparing a model.



Actual starting-shape chooser. Cross-section cards identify the outline; rotate the solid preview to inspect thickness and holes.

1. **Choose a design starting point**
   New project → Stock presets → A shape. All four blanks have height equal to 100% of the largest horizontal dimension. A has bilateral symmetry and a through-hole; heart is bilateral, cross has fourfold rotational and mirror symmetry, bow mirrors horizontally and vertically.

2. **Inspect before creating**
   Inspect the solid, then choose Use A-shaped blank · New project. It starts with zero CUTs; add a crown cut on the right. Canceling or going back preserves the original design.

3. **Compare before and after cutting**
   Try one cut and adjust depth while observing the A's hole and legs. Save and undo to compare outline retention at the same angle. Symmetric rough does not automatically mirror cutting instructions.

4. **Test your import workflow**
   Download the selected blank as OBJ and reimport through Upload blank. Keep units unspecified and +Z up; check the 100% height and through-hole. The default starting option still includes its table and girdle.

Pause and make a design judgment

Do the preset's recesses, holes and thickness suit your starting point in top and side views? Does one cut remove only what you intended?

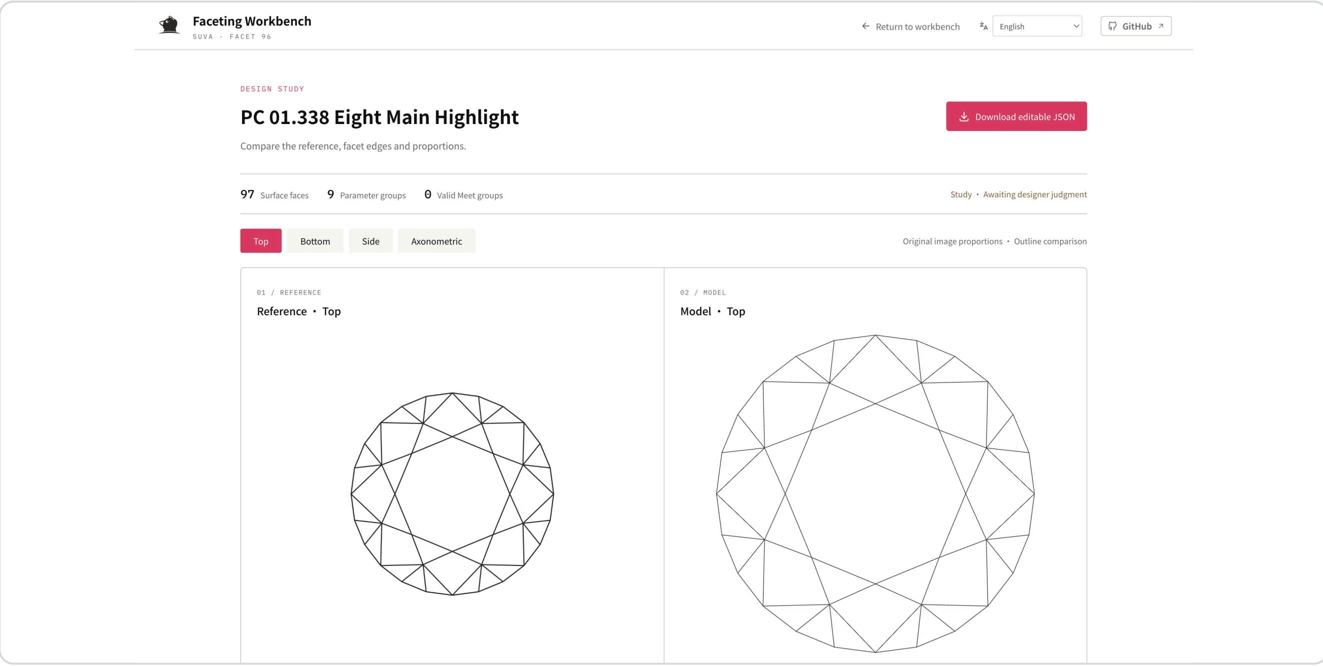
Before you continue

OBJ downloads in stock-presets/ use the same outlines as the built-in presets. A blank is a starting shape, not a finished gemstone cut.

CUT STUDY · CODEX ASSISTANCE

Turn a reference into an editable design

Bring a reference image and design intent. State the outline, table and girdle thickness to retain, then compare actual models and resolve unclear connections step by step.



Eight Main Highlight study page: the left reference is its technical top view; the right uses the same preset JSON. This demonstrates comparison workflow, not photographic reconstruction accuracy.

1. **Tell Codex your goal**
   Use facet-parametric-design in the workbench project with an image you have permission to use. Ask it to separate known details from inferences and deliver editable JSON with real-view comparisons. The workbench can edit and import independently without Codex.

2. **Compare under the same conditions**
   Open the generated study page and inspect top, bottom, side and axonometric views. Missing independent references are disclosed. Images must not be stretched to manufacture a match.

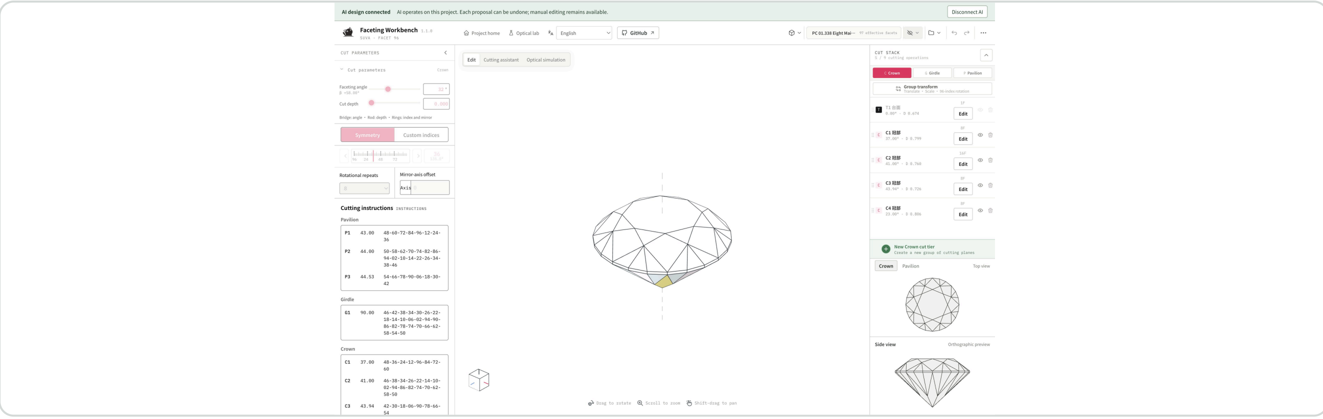
3. **Keep the study and refine it**
   Download editable JSON and import it through the file menu in a new project. Adjust one group’s angle or depth and inspect facet width and girdle thickness. Save the accepted variation and compare from the same angle.

> **Pause and make a design judgment**
> Which connection or proportion did this version resolve? Which areas still depend on inference and need a clearer reference or your judgment?

OPTIONAL LOCAL AI · CONVERSATIONAL DESIGN

Combine conversation and manual editing

Compare alternatives in conversation, then refine familiar parameters manually. Both use the same CUTs and undo history. The manual workbench requires neither Codex nor MCP.



Actual connected workbench: Eight Main Highlight, 97 effective facets. The following steps share one design between conversation and manual editing.

1. **Open the connected workbench**
   Ask an MCP-capable client to open the local workbench and follow its returned link. The top bar shows AI design connected. Ask the conversation to confirm the project name, then describe the outline to keep and the facets to change. Ordinary web use needs no connection.

2. **Compare before applying**
   Request real top, bottom, side and 3D previews and an explanation of affected facets. After applying, inspect the actual CUT STACK tiers; Undo reverts the change in one step. Save or discard a manual edit before continuing through conversation.

3. **Refine manually and archive**
   Edit the target tier, adjust angle or depth, compare from the same view and save. Request editable JSON in conversation or export it from File. Use JSON as a reliable transfer between browsers or local ports.

4. **Return to manual design at any time**
   Click Disconnect AI at the top. The project, history and parameter controls remain usable. Users without AI can create, import, edit, save and export in the standalone static workbench.

Pause and make a design judgment

At the same top and side views, did the crown change retain the intended outline and meetings? Which steps suit conversational exploration, and which need your manual judgment?

Before you continue

This page checks collaboration workflow, not cutting or optical gains. See docs/mcp/README.md for MCP installation and development. Codex and other local MCP clients are supported.